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Function for Hinge Loss for Multiple Points | Linear Algebra using Python

Linear Algebra using Python | Function for Hinge Loss for Multiple Points: Here, we are going to learn about the Function for hinge loss for multiple points and its implementation in Python.
Submitted by [Anuj Singh](#) on June 09, 2020

Hinge Loss is a loss function used in Machine Learning for training classifiers. The hinge loss is a maximum margin classification loss function and a major part of the SVM algorithm.

Hinge loss function is given by:

$$Loss_H = \max(0, (1 - Y * y))$$

Where, **Y** is the Label and

$$y = \theta . x$$

This is the general Hinge Loss function and in this tutorial, we are going to define a function for calculating the Hinge Loss for a multiple point (having one feature) with given **θ** .

Python code for hinge loss for multiple points

```
# Linear Algebra Learning Sequence
# Hinge loss for Multiple Point

import numpy as np

def hinge_loss_single(feature_vector, label, theta, theta_0):
    ydash = label*(np.matmul(theta,feature_vector) + theta_0)
    hinge = np.max([0.0, 1 - ydash*label])
    return hinge

def hinge_loss_full(feature_matrix, labels, theta, theta_0):
    tothinge = 0
    num = len(feature_matrix)
    for i in range(num):
        tothinge = tothinge + hinge_loss_single(feature_matrix[i], labels[i], theta, theta_0)

    hinge = tothinge

    return hinge

feature_matrix = np.array([[2,2], [3,3], [7,0], [14,47]])
theta = np.array([0.002,0.6])
theta_0 = 0
labels = np.array([[1], [-1], [1], [-1]])

hingell = hinge_loss_full(feature_matrix, labels, theta, theta_0)

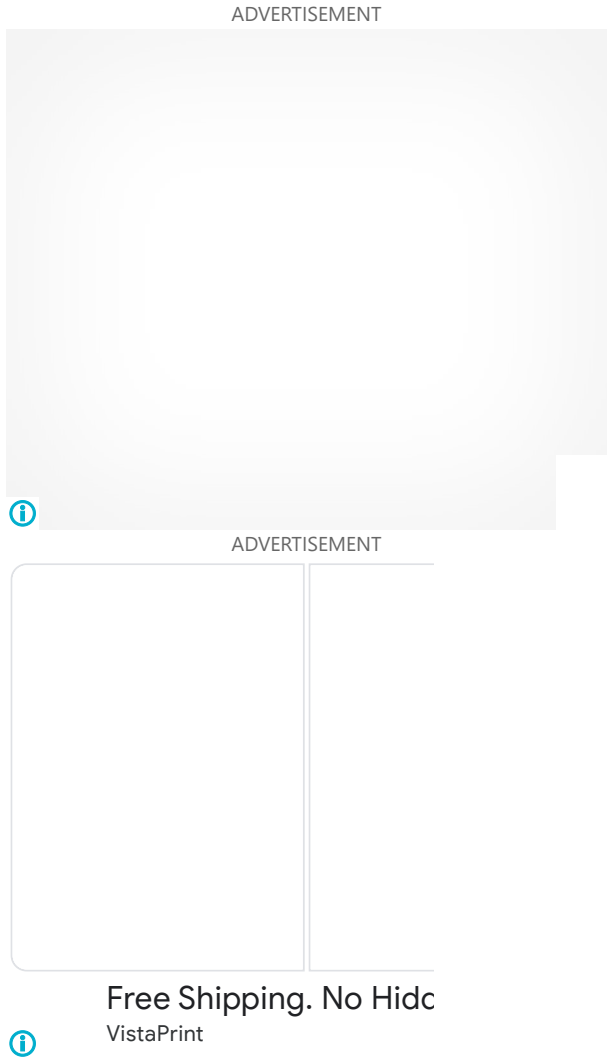
print('Data point: ', feature_matrix)
print('\n\nCorresponding Labels: ', labels)
print('\n\nHingle Loss for given data :', hingell)
```

Output:

```
Data point:  [[ 2  2]
 [ 3  3]
 [ 7  0]
 [14 47]]

Corresponding Labels:  [[ 1]
 [-1]
 [ 1]
 [-1]]

Hingle Loss for given data : [0.986]
```



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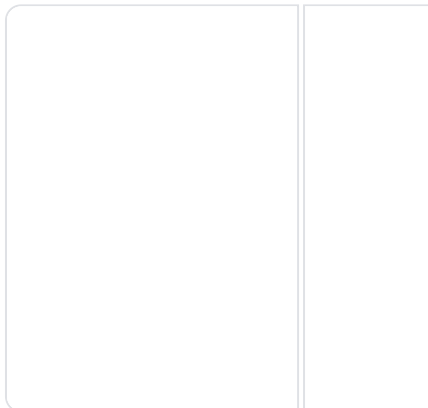
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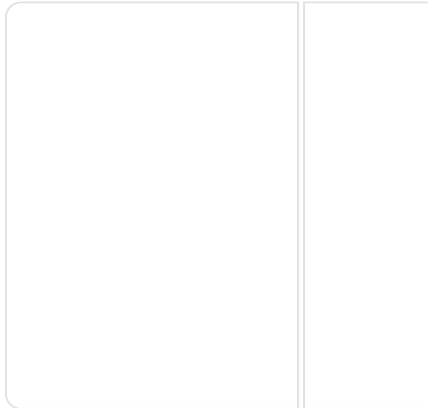
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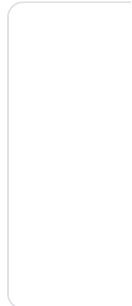


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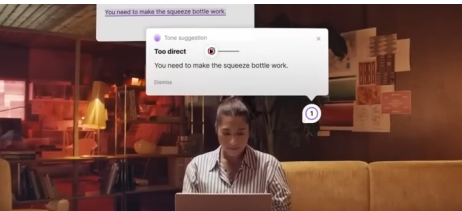
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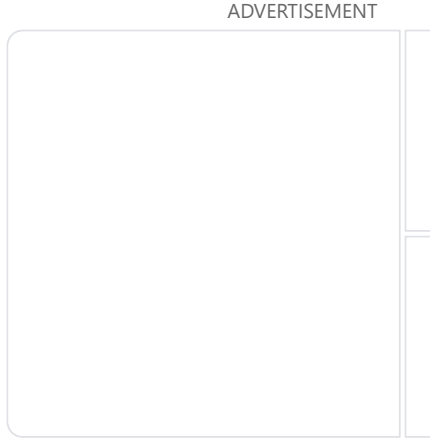
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- ✔ Sort an array of 0's, 1's and 2's in linear time complexity
- ✔ Checking Anagrams (check whether two string is anagrams or not)
- ✔ Relative sorting algorithm
- ✔ Finding subarray with given sum
- ✔ Find the level in a binary tree with given sum K
- ✔ Check whether a Binary Tree is BST (Binary Search Tree) or not
- ✔ 1[0]1 Pattern Count
- ✔ Capitalize first and last letter of each word in a line
- ✔ Print vertical sum of a binary tree
- ✔ Print Boundary Sum of a Binary Tree
- ✔ Reverse a single linked list
- ✔ Greedy Strategy to solve major algorithm problems
- ✔ Job sequencing problem
- ✔ Root to leaf Path Sum
- ✔ Exit Point in a Matrix
- ✔ Find length of loop in a linked list
- ✔ Toppers of Class
- ✔ Print All Nodes that don't have Sibling
- ✔ Transform to Sum Tree
- ✔ Shortest Source to Destination Path

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